Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2001-II-Q05] | Part A (Q1-Q5)

Answer: 253

1. If $a_1 < \cdots < a_{10}$ contains no triangle, then $a_{i+2} \geq a_i + a_{i+1}$. Starting with the least possible first terms $4, 5$, this forces the sequence at least $4, 5, 9, 14, 23, 37, 60, 97, 157, 254$.

2. Thus no triangle-free ten-set fits below 254. Conversely the displayed sequence has no triangle: any two entries before a_j sum to at most $a_{j-2} + a_{j-1} = a_j$.

3. Therefore $n = 253$ works and 254 fails, giving 253.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2004-I-Q05] | Part A (Q1-Q5)

Answer: 849

1. Alpha's ratios are $8/15 < 7/10$. Beta's ratio on each day is below $7/10$, so its total integer score is below 350, hence at most 349.

2. This is attainable: Beta scores 1 of 2 on day one and 348 of 498 on day two. These ratios are strictly below $8/15, 7/10$, and the first-day amount is not 300.

3. Thus the maximum ratio is $349/500$, giving 849.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2006-I-Q02] | Part A (Q1-Q5)

Answer: 901

1. The minimum is $1 + \cdots + 90 = 4095$, maximum $11 + \cdots + 100 = 4995$.

2. Unless the subset is the maximum final block, some selected integer $k < 100$ has unselected successor $k + 1$. Replacing it raises the sum by exactly one. Repetition constructs every intermediate sum.

3. Thus the number is $4995 - 4095 + 1 = 901$.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2007-II-Q08] | Part B (Q6-Q10)

Answer: 896

1. Let there be h length-four and v length-five lines. Then $4h + 5v = 2007$ and basic count $(h-1)(v-1)$. Put $x = h-1, y = v-1$, so $4x + 5y = 1998$.

2. The product $xy = x(1998 - 4x)/5$ is a concave quadratic, peaking at $x = 249.75$. Integer feasibility requires $x \equiv 2 \pmod 5$, whose nearest candidates are $247, 252$, yielding products $49894, 49896$.

3. The latter is maximal, with remainder 896.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2022-I-Q07] | Part B (Q6-Q10)

Answer: 289

1. The choice $(2 \cdot 3 \cdot 6 - 1 \cdot 5 \cdot 7)/(4 \cdot 8 \cdot 9) = 1/288$ provides an attainable bound.

2. Every positive numerator is an integer and every denominator is at most $7 \cdot 8 \cdot 9 = 504$. A numerator at least 2 gives a value at least $2/504 > 1/288$. A better value would therefore require numerator 1 and denominator greater than 288.

3. The denominator triples above 288 are $789, 689, 679, 589, 678, 579$ (each denotes a product). Their complementary digits always include both 2 and 4, so both triple products cannot have opposite parity unless one triple contains both; checking the ten unordered splits per complement gives minimum positive differences $6, 2, 2, 8, 6, 4$ respectively. None is 1. Thus the minimum is $1/288$, yielding 289.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2025-II-Q08] | Part B (Q6-Q10)

Answer: 610

1. An optimal choice has at most nine pennies and four dimes, since ten pennies can be replaced by a dime and five dimes by two quarters. Thus its non-quarter value is at most 49, so it uses either $q = \lfloor N/25 \rfloor$ or $q - 1$ quarters.

2. Let $r = N \bmod 25$ and $h(v) = \lfloor v/10 \rfloor + (v \bmod 10)$. Compare greedy count $q + h(r)$ with rival $q - 1 + h(r + 25)$. For $q \geq 1$, the rival wins exactly when $r \in \{5, 6, 7, 8, 9, 15, 16, 17, 18, 19\}$.

3. Each of these ten residue classes has forty representatives up to 1000, but its representative below 25 has no rival. Failures $= 10(40 - 1) = 390$, successes $= 610$.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2003-II-Q12] | Part C (Q11-Q15)

Answer: 134

1. With committee size t and candidate votes v, the condition is $100v/t \leq v - 1$, or $v(t - 100) \geq t$. Hence $t > 100$ and $v \geq \lceil t/(t - 100) \rceil$.

2. For $101 \leq t \leq 133$, the lower bound is at least five votes per candidate, forcing $t \geq 135$, a contradiction.

3. At $t = 134$, four votes suffice for each candidate. Give twenty-six candidates four votes and the last thirty; all satisfy the inequality. Thus the minimum is 134.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2009-II-Q12] | Part C (Q11-Q15)

Answer: 803

1. The sum of used elements is at least $1 + \cdots + 2k = k(2k+1)$. The distinct pair sums total at most $2009 + 2008 + \cdots + (2010 - k) = k(4019 - k)/2$. Thus $5k \leq 4017$, so $k \leq 803$.

2. Attain this with 401 pairs $(i, 1206 + i)$ for $1 \leq i \leq 401$, giving even sums $1208, \ldots, 2008$, and 402 pairs $(a, a + 403)$ for $402 \leq a \leq 803$, giving odd sums $1207, \ldots, 2009$.

3. Used elements are 1–803 and 805–1607, with no repeats, and all sums satisfy the rule. Maximum 803.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2022-II-Q14] | Part C (Q11-Q15)

Answer: 188

1. To form 1, necessarily $a = 1$. A collection of x ones, y stamps of value b, and z of value c covers a full interval through its total exactly when $x \geq b - 1$ and $x + yb \geq c - 1$; the total must also be at least 1000. These follow by preventing gaps at b and c.

2. If $c \leq 10$, even 97 stamps total at most 970, so $f \neq 97$. For $c = 11, b = 7$, six ones, one seven, and ninety elevens give 97 stamps. With only 96, at least six ones and one seven are required, and the total is at most $6 + 7 + 89 \cdot 11 = 992$. Hence $f = 97$.

3. For any $12 \leq c \leq 87$ and $2 \leq b < c$, use $b - 1$ ones and $c - b$ copies of b. These $c - 1$ stamps have value $b - 1 + b(c - b) \geq 2c - 3$. Adding $\lceil 1003/c \rceil - 2$ copies of c covers through 1000. The count is $c - 3 + \lceil 1003/c \rceil \leq 96$, since the concave quadratic $c(99 - c)$ is at least 1003 throughout this interval. Thus none of these c qualifies.

4. For $c = 88$ or 89, take $b = 87$, eighty-six ones, one 87, and ten copies of c: 97 stamps suffice. Only 96 stamps would total at most $86 + 87 + 9c < 1000$, because the smaller denominations are mandatory. Thus both have $f = 97$, and the answer is $11 + 88 + 89 = 188$.

Source / 来源

Section 6.2 Discrete extrema & constructions

离散最值与构造

[AIME-2026-I-Q11] | Part C (Q11-Q15)

Answer: 896

1. Orient each edge from its lower to its higher value. Writing $M = \sum c_v a_v$, the coefficient satisfies $|c_v| \le d_v$ and $\sum c_v = 0$, where degree d_v is four at 36 interior cells, three at 24 edge cells, two at four corners.

2. Therefore $M \le \sum d_v |a_v - 65/2|$. Rearrangement pairs the 36 largest deviations (values 1–$18, 47$–64, sum 828) with degree four, the next 24 (sum 192) with degree three, and the last four (sum 4) with degree two. Bound $M \le 4 \cdot 828 + 3 \cdot 192 + 2 \cdot 4 = 3896$.

3. Attain the bound using a checkerboard: all low values 1–32 on one color, all high values 33–64 on the other, allocating each degree class the indicated extremes. Each color has the required 18 interior,12 edge,2 corner cells. Thus the maximum is 3896, remainder 896.

Source / 来源